

CHAPTER 11

PROBABILITY

11.0 INTRODUCTION

In earlier chapters we have dealt with frequency distributions which refer to a sample of values (numerical data) drawn from a statistical population represented by classes together with the corresponding class frequencies. In this chapter we develop the mathematical theory of probability and introduce the concept of random variables which form the basis for various types of theoretical distributions.

11.1 PROBABILITY

Unit - III
(Definition. An experiment is defined as an action which we conceive and do or intend to do.)

All kinds of investigations do require repeated experiments with essentially same conditions with more or less same procedure. Each experiment ends with an *outcome*. For example, a research student in 'Statistics', when undertaking a pre election sample survey, conducts experiments with a sample of voters in the electoral roll following a well defined procedure adopted by himself and he gets an outcome.

In some experiments, the outcome *can* be predicted with certainty prior to the performance of the experiment. For example, in finding the areas of circles of different radii they can be predicted and calculated from the formula $A = \pi r^2$ for different values of r .

There are several experiments in which the outcome *cannot* be predicted with certainty but all possible outcomes of the experiment can be described and determined prior to the performance of the experiment. For

example in tossing a coin the outcome Head or Tail cannot be predicted with certainty but the outcome can be determined as Head or Tail. This leads to the concept of *random experiment*.

Definition. An experiment is called a **random experiment** if, when repeated under the same conditions, it is such that the outcome can not be predicted with certainty but all possible outcomes can be determined prior to the performance of the experiment.)

Each performance of the random experiment is called a **trial**. The collection of all possible outcomes of a random experiment is called the **sample space** and it is denoted by S . The elements of sample space are called **sample points**.)

Example 1. In the toss of a single coin let the outcome 'Head turning up' be denoted by H and the outcome 'Tail turning up' be denoted by T . The coin is repeatedly tossed under the same condition. The the toss of the coin results in the outcome H or T ; Thus yielding the sample space $S = \{H, T\}$.)

Example 2. When 2 coins are tossed at a time the outcome is an ordered pair (H, H) or (H, T) or (T, H) or (T, T) . Hence for the random experiment of tossing two coins, sample sapce is $S = \{(H, H), (H, T), (T, H), (T, T)\}$.)

Example 3. A die is numbered as 1, 2, 3, 4, 5, 6 on the faces. When this die is thrown the sample space is $S = \{1, 2, 3, 4, 5, 6\}$.)

Example 4. When two dice are thrown (or a die is thrown twice) then sample space is $S = \{(1, 1), (1, 2), \dots, (1, 6), (2, 1), (2, 2), \dots, (2, 6), \dots, \dots, \dots, \dots, (6, 1), (6, 2), \dots, (6, 6)\}$
 $= \{(x, y) \mid x, y \in N, 1 \leq x \leq 6, 1 \leq y \leq 6\}$ and $|S| = 36$.)

Example 5. When a die numbered as 1, 2, ..., 6 on its faces and a coin are tossed the sample space is $S = \{(1, H) (2, H) \dots (6, H), (1, T) (2, T) \dots (6, T)\}$ and $|S| = 12$.)

(Example 6. In the random experiment of drawing one card from a pack of 52 cards each outcome is any particular card and hence the sample space S consists of all individual cards and $|S| = 52$.)

(Example 7. A bulb is allowed to burn continuously till it expires. Then the sample point (outcome) e may be expressed in number of hours (minutes and seconds ignored) and $e > 0$. Then the sample space is $S = \{1, 2, 3, \dots\}$ which is an infinite set implying that a sample space may be finite or infinite.)

Definition. Any subset A of a sample space S is called an **event**.

The event S is called a **sure event** and the event Φ is called an **impossible event**.

Example 1. In example 2 given above

$A = \{(H, H)\}$ is the event of getting 2 heads.

$B = \{(H, T), (T, H)\}$ is the event of getting exactly one head. (a tail.)

Example 2. In example 3 given above

$A = \{1, 3, 5\}$ is the event of getting an odd number.

$B = \{2, 4, 6\}$ is the event of getting an even number.

Example 3. In example 4 given above

$A = \{(1, 4), (4, 1), (2, 3), (3, 2)\}$ is the event of getting the sum 5.

$B = \{(1, 1), (1, 2), (1, 3), (3, 1), (2, 1), (2, 2)\}$ is the event of getting the sum less than 5.

$C = \{(4, 6), (6, 4), (5, 5), (5, 6), (6, 5), (6, 6)\}$ is the event of getting the sum greater than 9.

Since events are subsets of S the concepts of sets and algebra of sets such as union, intersection, complements can be carried out to events of a random experiment. Also all the laws of set theory such as commutative, associative, distributive, De-Morgan's laws etc hold for algebra of events.

Illustration. Let A, B, C be any three events in a sample space S associated with a random experiment. Let $\bar{A}, \bar{B}, \bar{C}$ be the complement events. Then we may have the following occurrence of events and their related set theoretic expressions.

Sl. No.	events	set theoretic expressions
1.	Only A happens	$A \cap \bar{B} \cap \bar{C}$
2.	All the three events A, B, C happen	$A \cap B \cap C$
3.	None of the three events happen	$\bar{A} \cap \bar{B} \cap \bar{C}$ (or) $\overline{A \cup B \cup C}$
4.	A and B happen but not C	$A \cap B \cap \bar{C}$
5.	At least one of the 3 events happens	$A \cup B \cup C$
6.	Atleast 2 of the 3 events happens	$(\bar{A} \cap B \cap C) \cup (A \cap \bar{B} \cap C) \cup (A \cap B \cap \bar{C})$
7.	Exactly one event occurs	$(A \cap \bar{B} \cap \bar{C}) \cup (\bar{A} \cap B \cap \bar{C}) \cup (\bar{A} \cap \bar{B} \cap C)$

Definition. Let S be a sample space associated with an experiment. Let A be an event. Suppose an experiment is repeated N times and suppose the event A happens f times. Then f/N is called the **relative frequency** of the event A . Clearly $0 \leq f/N \leq 1$.

For example, in the sample space of throwing two dice $S = \{(x, y) \mid x, y \in \mathbb{N}, 1 \leq x \leq 6, 1 \leq y \leq 6\}$ consider an event $A = \{(1, 6), (2, 5), (3, 4), (4, 3), (6, 1)\}$ which consists of the sample points (a, b) for which $a + b = 7$. Suppose that the die is thrown 400 times. Hence $N = 400$. Suppose that the number of times the sum seven occurs is 80 then

$f = 80$ and the relative frequency with which the outcome was in A is $f/N = 80/400 = 0.2$.

The probability $P(A)$ of the event A is that number about which the relative frequency f/N of the event tends to stabilize after a long series of experiments. This suggests the following desirable properties to be satisfied by a probability function P .

- (a) $P(A) \geq 0$ for all events.
- (b) $P(A) \leq 1$ for all events.
- (c) $P(S) = 1$.
- (d) If A and B are disjoint events $P(A \cup B) = P(A) + P(B)$.

This motivates the following formal definition of a probability set function.

Definition. Let S be a given sample space. Suppose a real number $P(A)$ is assigned to certain subsets of S and the set function P satisfies the following conditions. $C_1: P(A) \geq 0$ for all $A \subseteq S$.

$$C_2: P(S) = 1.$$

$C_3: \text{If } \{A_n\} \text{ is any finite or infinite sequence of disjoint events then } P(\bigcup_i A_i) = \sum_i P(A_i)$

Then P is called the **probability set function** and the number $P(A)$ is called the **probability** of the event A .

Example. Let $|S| = N$ and $P(\{a_i\}) = \frac{1}{N}$ for each $a_i \in S$. Then clearly

P defines a probability (set) function on S and P is given by $P(A) = \frac{|A|}{N}$

for every $A \subseteq S$. P is known as the **uniform probability function**. This concept of uniform probability coincides with the classical definition of

probability, namely $P(A) = \frac{\text{Number of cases favourable to } A}{\text{Total number of cases}}$.

The following theorems give some other properties of probability

(set) function.

Theorem 11.1 Let S be a sample space. Let $A \subseteq S$. Then

(i) $P(A) = 1 - P(\bar{A})$ where $\bar{A}$ is the complement of A

(ii) $P(\Phi) = 0$

Proof. (i) We have $S = A \cup \bar{A}$ and $A \cap \bar{A} = \Phi$.

Hence by C_3 , $P(S) = P(A) + P(\bar{A})$

$\therefore 1 = P(A) + P(\bar{A})$ (by C_2)

$\therefore P(A) = 1 - P(\bar{A})$.

(ii) Take $A = \Phi$ in (i).

Then $\bar{A} = S$ and $P(\Phi) = 1 - P(S) = 1 - 1 = 0$.

Theorem 11.2. If A and B are events of a sample space S such that $A \subseteq B$ then $P(A) \leq P(B)$.

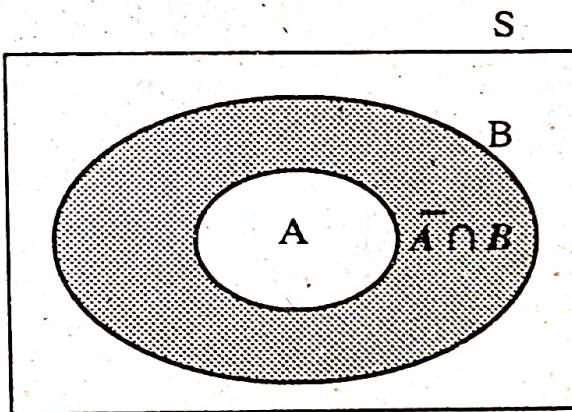
Proof. Let $A \subseteq B$. Then

$B = A \cup (\bar{A} \cap B)$ and A and $\bar{A} \cap B$ are disjoint events of S .

Hence $P(B) = P(A) + P(\bar{A} \cap B)$

But $P(\bar{A} \cap B) \geq 0$ (by C_1). Hence

$P(B) \geq P(A)$.



Corollary. For each $A \subseteq S$, $0 \leq P(A) \leq 1$.

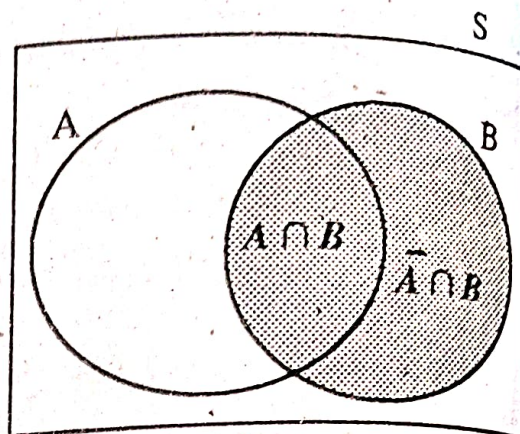
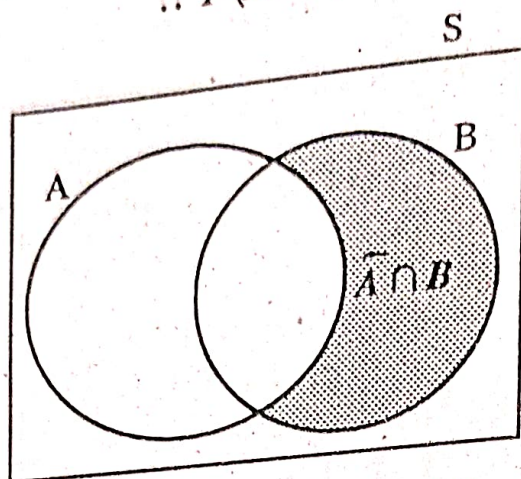
Proof. $\Phi \subseteq A \subseteq S$. Hence $P(\Phi) \leq P(A) \leq P(S)$. Hence $0 \leq P(A) \leq 1$.

Theorem 11.3 If A and B are any two events of a sample space S then

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Proof. $A \cup B = A \cup (\bar{A} \cap B)$ and A and $\bar{A} \cap B$ are disjoint sets.

$$\therefore P(A \cup B) = P(A) + P(\bar{A} \cap B) \quad \dots\dots (1)$$



Now, $B = (A \cap B) \cup (\bar{A} \cap B)$ and $A \cap B$ and $\bar{A} \cap B$ are disjoint sets.

$$\therefore P(B) = P(A \cap B) + P(\bar{A} \cap B)$$

$$\therefore P(\bar{A} \cap B) = P(B) - P(A \cap B) \quad \dots\dots (2)$$

Substituting (2) in (1) we get, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Note. The above theorem can be generalised for any n events $A_1, A_2, \dots, A_n$ of a sample space.

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i) - \sum_{i,j}^{n,n} P(A_i \cap A_j) + \sum_{i,j,k}^{n,n,n} P(A_i \cap A_j \cap A_k) - \dots\dots (-1)^{n-1} P(A_1 \cap A_2 \cap \dots \cap A_n).$$

Example 1. Let $S = \{(H, H), (H, T), (T, H), (T, T)\}$. Let us assign a probability of $1/4$ to each element of the sample space S .

Let $A = \{(H, H), (H, T)\}$; $B = \{(H, H), (T, H)\}$.

Then $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{2}$, $P(A \cup B) = \frac{3}{4}$ and $P(A \cap B) = \frac{1}{4}$ and

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{4}$ and the theorem 11.3 is verified.

Example 2. Let $S = \{(i, j)/i, j \in N, 1 \leq i \leq 6, 1 \leq j \leq 6\}$ be the sample space of the random experiment of throwing two dice. We assign the uniform probability of $1/36$ to each of the 36 sample points in the sample space S .

Let A denote the event of getting 1 in the second die. Then $A = \{(1, 1), (2, 1), (3, 1), (4, 1), (5, 1), (6, 1)\}$ so that $P(A) = 6/36 = 1/6$.
Let $B = \{(1, 2), (2, 2), (3, 2)\}$. Then $P(B) = 3/36 = 1/12$.

Now $A \cap B = \Phi$ we note that $P(A \cup B) = \frac{9}{36} = \frac{1}{4} = P(A) + P(B)$.

Definition. Let S denote a sample space associated with an experiment. Let $A_1, A_2, \dots, A_n, \dots$ be a sequence of subsets of S .

If $A_i \cap A_j = \Phi$ for all i, j with $i \neq j$ then the sequence of subsets is said to be **mutually disjoint**.

If $\bigcup_{n=1}^{\infty} A_n = S$ then the sequence of events is said to be **exhaustive**.

Note. If the sequence of sets $A_1, A_2, \dots, A_n, \dots$ is mutually disjoint and

exhaustive then $P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n) = 1$.

Example 1. Let $S = \{(i, j)/i, j \in N, 1 \leq i \leq 6, 1 \leq j \leq 6\}$. Let A be an event of getting the sum $i + j$ and odd number as B be an event of getting the sum as even number. (i.e.) $A = \{(i, j)/(i, j) \in S \text{ and } i + j \text{ is odd}\}$ and $B = \{(i, j)/(i, j) \in S \text{ and } i + j \text{ is even}\}$.

Clearly $A \cap B = \Phi$ and $A \cup B = S$. Hence A and B are mutually exclusive and exhaustive events.

In the example 2 given above A and B are mutually exclusive events but not exhaustive.

11.2 CONDITIONAL PROBABILITY

Definition. Let S be a sample space with the probability function P defined on it. Let A and B be two events of S with $P(B) > 0$. Then the conditional probability of A given B , written as $P(A|B)$ is defined as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

Remarks . 1. $P(A|B) \geq 0$

2. $P(A|A) = 1$

3. $P(A_1 \cup A_2 \cup \dots | B) = P(A_1|B) + P(A_2|B) + \dots$ provided $A_1, A_2, \dots$ are mutually disjoint events.

Note. From the definition of conditional probability for any two events A and B we have $P(A \cap B) = P(B) P(A|B)$. This relation is called multiplication theorem for probabilities.

Theorem 11.4. Let $A_1, A_2, \dots, A_n$ be n events in a sample space S . Then

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) P(A_2|A_1) P(A_3|A_1 \cap A_2) \dots \\ \dots P(A_n|A_1 \cap \dots \cap A_{n-1})$$

Proof. $P(A_1) P(A_2|A_1) P(A_3|A_1 \cap A_2) \dots P(A_n|A_1 \cap A_2 \cap \dots \cap A_{n-1})$

$$= P(A_1) \cdot \frac{P(A_1 \cap A_2)}{P(A_1)} \cdot \frac{P(A_1 \cap A_2 \cap A_3)}{P(A_1 \cap A_2)} \dots \frac{P(A_1 \cap A_2 \cap \dots \cap A_n)}{P(A_1 \cap A_2 \cap \dots \cap A_{n-1})}$$

$$= P(A_1 \cap A_2 \cap \dots \cap A_n).$$

Intuitively let A and B be two events. The event A is said to be independent of the event B if the occurrence of B does not affect the occurrence of A . This is formulated mathematically as follows.

Definition. Let A and B be two events with $P(B) \neq 0$. A is said to be independent of B if $P(A|B) = P(A)$.

Remark. If two events A and B are such that $P(A) \neq 0$ and $P(B) \neq 0$ and if A is independent of B then B is independent of A and hence we say that A and B are independent.

Proof. A is independent of B .

$$\Rightarrow P(A|B) = P(A)$$

$$\Rightarrow \frac{P(A \cap B)}{P(B)} = P(A)$$

$$\Rightarrow P(A \cap B) = P(A) P(B)$$

$$\Rightarrow P(B \cap A) = P(A) P(B)$$

$$\Rightarrow \frac{P(B \cap A)}{P(A)} = P(B)$$

$$\Rightarrow P(B|A) = P(B).$$

$$\Rightarrow B \text{ is independent of } A.$$

Note. If A and B are two independent events then $P(A \cap B) = P(A) P(B)$.

Definition. A set of events $A_1, A_2, \dots, A_n$ are said to be **pairwise independent** if $P(A_i \cap A_j) = P(A_i) P(A_j)$ for all $i \neq j$.

Definition. The events $A_1, A_2, \dots, A_n$ are said to be **mutually independent**

if $P(A_i \cap A_j) = P(A_i) P(A_j)$ for all $i \neq j$ and $i, j = 1, 2, \dots, n$ and

$P(A_i \cap A_j \cap A_k) = P(A_i) P(A_j) P(A_k)$ for all $i \neq j \neq k$ and $i, j, k = 1, 2, \dots, n$.

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) P(A_2) \dots P(A_n).$$

Definition. The events $A_1, A_2, \dots, A_n$ are said to be **independent** if

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) P(A_2) \dots P(A_n)$$

Note. Pairwise independence of n events $\nRightarrow$ independence of n events.

For example consider the random experiment of tossing 2 coins.
Then the sample space is $S = \{(H, H), (H, T), (T, H), (T, T)\}$.

Let $A_1 = \{(H, H), (H, T)\}$; $A_2 = \{(H, H), (T, H)\}$ and $A_3 = \{(H, T), (T, H)\}$

$$P(A_1) = P(A_2) = P(A_3) = 1/2$$

$$P(A_1 \cap A_2) = P(\{(H, H)\}) = 1/4 = P(A_1) P(A_2)$$

$$P(A_2 \cap A_3) = P(\{(T, H)\}) = 1/4 = P(A_2) P(A_3)$$

$$P(A_3 \cap A_1) = P(\{(H, T)\}) = 1/4 = P(A_3) P(A_1)$$

$\therefore A_1, A_2, A_3$ are pairwise independent events.

Now, $P(A_1 \cap A_2 \cap A_3) = P(\Phi) = 0$ and $P(A_1) P(A_2) P(A_3) = 1/8$

$$\therefore P(A_1 \cap A_2 \cap A_3) \neq P(A_1) P(A_2) P(A_3)$$

$\therefore A_1, A_2, A_3$ are not independent events.

Note. If A and B are two mutually exclusive events and $P(A) > 0$, $P(B) > 0$ then A and B are not independent events.

For, $P(A \cap B) = P(\Phi) = 0$ whereas $P(A) P(B) > 0$ so that $P(A \cap B) \neq P(A) P(B)$

Properties of independent events

Theorem 11.5. If A and B are independent events then A and $\bar{B}$ are also independent events.

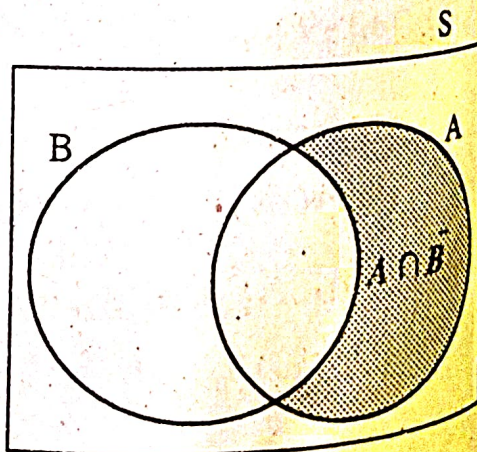
Proof. $A \cap \bar{B}$ and $A \cap B$ are disjoint events and

$$A = (A \cap \bar{B}) \cup (A \cap B).$$

$$\therefore P(A) = P(A \cap \bar{B}) + P(A \cap B)$$

Hence

$$P(A \cap \bar{B}) = P(A) - P(A \cap B)$$



$$\begin{aligned}
 &= P(A) - P(A) P(B) \text{ (since } A \text{ and } B \text{ are independent events)} \\
 &= P(A) [1 - P(B)] \\
 &= P(A) P(\bar{B}).
 \end{aligned}$$

Hence A and $\bar{B}$ are independent events.

Theorem 11.6. If A and B are independent events then $\bar{A}$ and $\bar{B}$ are also independent events.

Proof. Since A and B are independent $P(A \cap B) = P(A) P(B)$ (1)

$$\begin{aligned}
 \text{Now, } P(\bar{A} \cap \bar{B}) &= P(\overline{A \cup B}) \\
 &= 1 - P(A \cup B) \\
 &= 1 - [P(A) + P(B) - P(A \cap B)] \\
 &= 1 - P(A) - P(B) + P(A) P(B) \text{ (using (1))} \\
 &= [1 - P(A)] [1 - P(B)] \\
 &= P(\bar{A}) P(\bar{B}).
 \end{aligned}$$

Hence $\bar{A}$ and $\bar{B}$ are independent events.

Theorem 11.7. If A, B, C are mutually independent events, then $A \cup B$ and C are independent events.

$$\begin{aligned}
 \text{Proof. } P[(A \cup B) \cap C] &= P[(A \cap C) \cup (B \cap C)] \\
 &= P(A \cap C) + P(B \cap C) - P(A \cap B \cap C) \\
 &= P(A) P(C) + P(B) P(C) - P(A) P(B) P(C) \\
 &\quad \text{(since } A, B, C \text{ are mutually independent).} \\
 &= P(C) [P(A) + P(B) - P(A) P(B)] \\
 &= P(C) P(A \cup B) \\
 &= P(A \cup B) P(C)
 \end{aligned}$$

$\therefore A \cup B$ and C are independent events.

Theorem 11.8 If A, B, C are any three events in a sample space and if A, B, C are pairwise independent and A is independent of $B \cup C$ then A, B, C are mutually independent.

Proof. Since A, B, C are pairwise independent events, we have

$$P(A \cap B) = P(A) P(B)$$

$$P(B \cap C) = P(B) P(C)$$

$$P(A \cap C) = P(A) P(C)$$

..... (1)

Since A is independent of $B \cup C$ we have

$$P[A \cap (B \cup C)] = P(A) P(B \cup C)$$

..... (2)

$$\text{Now, } P[A \cap (B \cup C)] = P[(A \cap B) \cup (A \cap C)]$$

$$= P(A \cap B) + P(A \cap C) - P(A \cap B \cap C)$$

Hence, from (1) and (2) we have

$$P(A) P(B \cup C) = P(A) P(B) + P(A) P(C) - P(A \cap B \cap C)$$

$$\therefore P(A) [P(B) + P(C) - P(B \cap C)]$$

$$= P(A) P(B) + P(A) P(C) - P(A \cap B \cap C)$$

$$\therefore P(A) [P(B) + P(C) - P(B) P(C)] = P(A) P(B) + P(A) P(C) - P(A \cap B \cap C)$$

$$\therefore P(A \cap B \cap C) = P(A) P(B) P(C)$$

..... (3)

From (1) and (3) we get, A, B and C are mutually independent events.

Note. Let A, B be two events with $P(A) > 0$ and $P(B) > 0$. Then

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \dots\dots (1) \text{ and } P(B|A) = \frac{P(A \cap B)}{P(A)} \dots\dots (2).$$

Substituting (2) in (1) we get, $P(A|B) = \frac{P(A) P(B|A)}{P(B)}$.

This is known as Baye's formula.

Theorem 11.9. (Baye's Theorem) Let $\{A_i\}$ be a sequence of mutually exclusive and exhaustive events in a sample space S such that $P(A_i) > 0$ for all i . Let B be any event with $P(B) > 0$. Then $P(A_i|B) = \frac{P(A_i) P(B|A_i)}{\sum_i P(A_i) P(B|A_i)}$

Proof. By Baye's rule we have $P(A_i|B) = \frac{P(A_i) P(B|A_i)}{P(B)} \dots\dots (1)$

We claim that $P(B) = \sum_i P(A_i) P(B|A_i)$.

Since the events $\{A_i\}$ are mutually exclusive and exhaustive we have $\bigcup_i A_i = S$ and A_i^c are disjoint.

$$\begin{aligned} \therefore B &= B \cap S = B \cap \left(\bigcup_i A_i \right) \\ &= \bigcup_i (B \cap A_i). \end{aligned}$$

$$\begin{aligned} \therefore P(B) &= P\left(\bigcup_i (B \cap A_i) \right) \\ &= \sum_i P(B \cap A_i) \\ &= \sum_i P(A_i) P(B|A_i) \end{aligned}$$

$$\text{Thus } P(B) = \sum_i P(A_i) P(B|A_i) \dots\dots\dots (2)$$

Substituting (2) in (1) we get the required result.

Solved problems

Problem 1. Two dice numbered as 1, 2, 3, 4, 5, 6 on the faces are thrown. Let A be the event that the sum of the points on the faces shown is odd and B is the event that at least one number is 1. Find the probabilities of the following (i) $A \cap B$ (ii) $A \cup B$ (iii) $A \cap \bar{B}$ (iv) $\bar{A} \cap \bar{B}$ (v) $A|B$ (vi) $B|A$.

Solution. The sample space is

$S = \{(x, y) \mid x, y \in \mathbb{N}, 1 \leq x \leq 6 \text{ and } 1 \leq y \leq 6\}$, we have $|S| = 36$.

$A = \{(1, 2), (1, 4), (1, 6), (2, 1), (2, 3), (2, 5), (3, 2), (3, 4), (3, 6), (4, 1), (4, 3), (4, 5), (5, 2), (5, 4), (5, 6), (6, 1), (6, 3), (6, 5)\}$

$B = \{(1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (3, 1), (4, 1), (5, 1), (6, 1), (1, 1)\}$

$A \cap B = \{(1, 2), (1, 4), (1, 6), (2, 1), (4, 1), (6, 1)\}$

$$\text{Now, } P(A) = \frac{|A|}{|S|} = \frac{18}{36} = \frac{1}{2} \text{ and } P(B) = \frac{|B|}{|S|} = \frac{11}{36}$$

$$(i) P(A \cap B) = \frac{|A \cap B|}{|S|} = \frac{6}{36} = \frac{1}{6}$$

$$(ii) P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{2} + \frac{11}{36} - \frac{1}{6} = \frac{23}{36}$$

$$(iii) P(A \cap \bar{B}) = P(A) - P(A \cap B) = \frac{1}{2} - \frac{1}{6} = \frac{1}{3}$$

$$(iv) P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) = 1 - P(A \cup B) = 1 - \frac{23}{36} = \frac{13}{36}$$

$$(v) P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{1/6}{11/36} = \frac{6}{11}$$

$$(vi) P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{1/6}{1/2} = \frac{1}{3}$$

Problem 2. A coin is tossed as many times as is necessary to turn up head. Thus the elements of the sample space S are $H, TH, TTH, TTTH, TTTTH$, etc... We assign to these elements the respective probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}$ etc. Show that this defines a probability set function P . Let

$A = \{H, TH, TTH, TTTH, TTTTH\}$ & $B = \{TTTTH, TTTTTH\}$.

Compute (i) $P(A)$ (ii) $P(B)$ (iii) $P(A \cap B)$ (iv) $P(A \cup B)$.

Solution. (i) To prove that P is a probability set function, it is enough to prove that $P(S) = 1$.

$$P(S) = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$$

$$= \frac{1}{2} \left[1 + \frac{1}{2} + \frac{1}{4} + \dots \right] = \frac{1}{2} \left[\frac{1}{1 - 1/2} \right] = 1.$$

$$(ii) P(A) = P(\{H\}) + P(\{TH\}) + P(\{TTH\}) + P(\{TTTH\}) + P(\{TTTTH\})$$

$$= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} = \frac{31}{32}.$$

$$(iii) P(B) = P(\{TTTTTH\}) + P(\{TTTTTTH\})$$

$$= \frac{1}{32} + \frac{1}{64} = \frac{3}{64}.$$

$$(iv) P(A \cap B) = P(\{TTTTTH\}) = \frac{1}{32}.$$

$$(v) P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{31}{32} + \frac{3}{64} - \frac{1}{32} = \frac{63}{64}.$$

Problem 3. A random experiment consists in drawing a card from an ordinary pack of 52 playing cards. Let the probability set function P assign a probability of $1/52$ to each of the 52 possible outcomes. Let A denote the collection of 13 hearts and let B denote the collection of the 4 kings. Compute

(i) $P(A)$ (ii) $P(B)$ (iii) $P(A \cap B)$ (iv) $P(A \cup B)$.

Solution. We have $|S| = 52$, $|A| = 13$, $|B| = 4$ and $|A \cap B| = 1$.

$$\text{Hence } P(A) = \frac{13}{52} = \frac{1}{4}; P(B) = \frac{4}{52} = \frac{1}{13}; P(A \cap B) = \frac{1}{52}$$

$$\text{Hence } P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{1}{4} + \frac{1}{13} - \frac{1}{52} = \frac{4}{13}.$$

Problem 4. If A and B are two events with $P(A) = 0.4$, $P(B) = 0.3$ and

$P(A \cap B) = 0.2$. Find the probability of the events (i) $A \cup B$ (ii) $\bar{A} \cup \bar{B}$

(iii) $\bar{A} \cap \bar{B}$.

Solution. (i) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $= 0.4 + 0.3 - 0.2 = 0.5$

(ii) $P(\overline{A \cap B}) = P(\overline{A \cap B})$
 $= 1 - P(A \cap B) = 1 - 0.2 = 0.8.$

(iii) The event B can be written as $B = (\overline{A} \cap B) \cup (A \cap B)$ and $\overline{A} \cap B$ and $A \cap B$ are disjoint events.

$$\therefore P(B) = P(\overline{A} \cap B) + P(A \cap B)$$

$$\therefore P(\overline{A} \cap B) = P(B) - P(A \cap B) = 0.3 - 0.2 = 0.1$$

Problem 5. Let A and B be two events such that $P(A) = 2/3$ and $P(B) = 5/6$. Show that (i) $P(A \cup B) \geq 2/3$ (ii) $1/2 \leq P(A \cap B) \leq 5/6$.

Solution. (i) Since $A \subseteq A \cup B$ we have $P(A) \leq P(A \cup B)$.

$$\text{Hence } P(A \cup B) \geq 2/3.$$

(ii) Since $A \cap B \subseteq B$, we have $P(A \cap B) \leq P(B)$.

$$\text{Hence } P(A \cap B) \leq 5/6 \quad \dots\dots (1)$$

$$\text{Also } P(A \cup B) \leq 1.$$

$$\therefore P(A) + P(B) - P(A \cap B) \leq 1$$

$$P(A) + P(B) - 1 \leq P(A \cap B)$$

$$\therefore \frac{2}{3} + \frac{5}{6} - 1 \leq P(A \cap B)$$

$$\therefore 1/2 \leq P(A \cap B) \quad \dots\dots (2)$$

$$\text{From (1) and (2) we get } 1/2 \leq P(A \cap B) \leq 5/6$$

Problem 6. The probabilities of n independent events are $p_1, p_2, \dots, p_n$.

Find the probability that at least one of the events happen.

Solution. Let the n independent events be $A_1, A_2, \dots, A_n$.

$$\text{Given } P(A_1) = p_1; P(A_2) = p_2; \dots, P(A_n) = p_n.$$

$$\therefore P(\bar{A}_1) = 1 - p_1; P(\bar{A}_2) = 1 - p_2; \dots P(\bar{A}_n) = 1 - p_n$$

$$\text{We have } P(A_1 \cup A_2 \cup \dots \cup A_n) + P(\overline{A_1 \cup A_2 \cup \dots \cup A_n}) = 1$$

$$\begin{aligned} \therefore P(A_1 \cup A_2 \dots \cup A_n) &= 1 - P(\bar{A}_1 \cap \bar{A}_2 \cap \dots \cap \bar{A}_n) \\ &= 1 - P(\bar{A}_1) P(\bar{A}_2) \dots P(\bar{A}_n) \end{aligned}$$

(since $A_1, A_2 \dots A_n$ are independent $\Rightarrow \bar{A}_1, \bar{A}_2 \dots \bar{A}_n$ are independent).
 $= 1 - (1 - p_1)(1 - p_2) \dots (1 - p_n)$. Hence the result.

Problem 7. (Boole's inequality). If A and B are any two events in a sample space then prove that (i) $P(A \cap B) \geq 1 - P(\bar{A}) - P(\bar{B})$ (ii) (Generalised Boole's inequality.) For the events $A_1, A_2, \dots, A_n, \dots$ in a sample space

$$P\left(\bigcap_{i=1}^{\infty} A_i\right) \geq 1 - \sum_{i=1}^{\infty} P(\bar{A}_i).$$

Solution. (i) For any two events A and B we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cap B) = [1 - P(\bar{A})] + [1 - P(\bar{B})] - P(A \cup B)$$

$$= 1 - P(\bar{A}) - P(\bar{B}) + [1 - P(A \cup B)]$$

$$\geq 1 - P(\bar{A}) - P(\bar{B}) \quad (\text{since } 0 \leq P(A \cup B) \leq 1)$$

(ii) Let $B_1 = \bigcap_{i=2}^{\infty} A_i$ so that $\bigcap_{i=1}^{\infty} A_i = A_1 \cap B_1$.

$$\text{Now, } P\left(\bigcap_{i=1}^{\infty} A_i\right) = P(A_1 \cap B_1) \geq 1 - P(\bar{A}_1) - P(\bar{B}_1) \quad (\text{by (i)})$$

$$= 1 - P(\bar{A}_1) - P\left(\overline{\bigcap_{i=2}^{\infty} A_i}\right)$$

$$= 1 - P(\bar{A}_1) - P\left(\bigcup_{i=2}^{\infty} \bar{A}_i\right) \geq 1 - P(\bar{A}_1) - \sum_{i=2}^{\infty} P(\bar{A}_i)$$

$$= 1 - \sum_{i=1}^{\infty} P(\bar{A}_i)$$

$$\therefore P\left(\bigcap_{i=1}^{\infty} A_i\right) \geq 1 - \sum_{i=1}^{\infty} P(\bar{A}_i)$$

Problem 8. If A and B are two events such that $P(A) = 1/3$ and $P(B) = 3/4$ and $P(A \cup B) = 11/12$, find $P(A|B)$ and $P(B|A)$.

Solution. For any two events A and B we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 1/3 + 3/4 - 11/12 = 1/6.$$

$$\text{Now, } P(A|B) = \frac{P(A \cap B)}{P(B)} = 1/6 \times 4/3 = 2/9.$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{1}{6} \times 3 = \frac{1}{2}.$$

Problem 9. From a pack of 52 cards two cards are drawn the first being replaced before the second is drawn. Find the probability that the first is a diamond and second is a king.

Solution. Let A_1 be an event of drawing a diamond.

$$\therefore P(A_1) = \frac{{}^{13}C_1}{{}^{52}C_1} = \frac{13}{52} = \frac{1}{4}.$$

Let A_2 be an event of drawing a king. Hence $P(A_2) = \frac{{}^4C_1}{{}^{52}C_1} = \frac{4}{52} = \frac{1}{13}.$

These two events are independent and hence

$$P(A_1 \cap A_2) = P(A_1) P(A_2) = 1/4 \times 1/13 = 1/52$$

Problem 10. Among the workers in a factory only 30% receive a bonus. Among those receiving the bonus only 20% are skilled. What is the probability of a randomly selected worker who is skilled and receiving bonus.

Solution. Let A denote the event of receiving bonus and let B denote the event of considering skilled workers.

$$\text{Given } P(A) = 0.3 \text{ and } P(B|A) = 0.2$$

We have to find the probability of the event $A \cap B$.

$$P(A \cap B) = P(A) P(B|A) = 0.3 \times 0.2 = 0.06.$$

Problem 11. X can hit a target 3 times in 5 shots; Y can hit 2 times in 5 shots; Z can hit 3 times in 4 shots. They fire a volley. What is the probability that 2 shots hit the target?

Solution. Let A be the event of X hitting the target;

B be the event of Y hitting the target;

C be the event of Z hitting the target.

$$\text{Given } P(A) = \frac{3}{5}; P(B) = \frac{2}{5}; P(C) = \frac{3}{4}$$

The event of two shots hitting the target is represented by

$$(A \cap B \cap \bar{C}) \cup (A \cap \bar{B} \cap C) \cup (\bar{A} \cap B \cap C).$$

Also A, B, C are independent events.

Hence the required probability

$$\begin{aligned} &= P(A \cap B \cap \bar{C}) + P(A \cap \bar{B} \cap C) + P(\bar{A} \cap B \cap C) \\ &= P(A) P(B) P(\bar{C}) + P(A) P(\bar{B}) P(C) + P(\bar{A}) P(B) P(C) \\ &= \frac{3}{5} \times \frac{2}{5} \times \left(1 - \frac{3}{4}\right) + \frac{3}{5} \times \left(1 - \frac{2}{5}\right) \times \frac{3}{4} + \left(1 - \frac{3}{5}\right) \times \frac{2}{5} \times \frac{3}{4} \\ &= \frac{9}{20}. \end{aligned}$$

Problem 12. The chance of X solving a particular problem is $\frac{2}{3}$ and solving the problem is $\frac{3}{4}$. What is the probability that the problem is solv

if they try independently.

Solution. Let A denote the event that X solves the problem and B denote the event that Y solves the problem.

$\therefore P(A) = \frac{2}{3}$ and $P(B) = \frac{3}{4}$ so that $P(\bar{A}) = \frac{1}{3}$ and $P(\bar{B}) = \frac{1}{4}$.

The required probability = $P(A \cup B)$

$$= P(A) + P(B) - P(A \cap B)$$

$$= P(A) + P(B) - P(A)P(B)$$

$$= \frac{2}{3} + \frac{3}{4} - \frac{1}{2} = \frac{11}{12}.$$

Problem 13. The chances that 4 students A, B, C, D solve a problem are $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{4}$ respectively. If all of them try to solve the problem, what is the probability that the problem is solved.

Solution. Let A_1 denote the event that A solves the problem.

A_2 denote the event that B solves the problem.

A_3 denote the event that C solves the problem.

A_4 denote the event that D solves the problem.

Given $P(A_1) = \frac{1}{2}; P(A_2) = \frac{1}{3}; P(A_3) = \frac{1}{4}; P(A_4) = \frac{1}{4}$. We

note that A_1, A_2, A_3, A_4 are independent events. The problem will be solved if atleast one of them solves it.

We have to find $P(A_1 \cup A_2 \cup A_3 \cup A_4)$. We note that probability of atleast one solves the problem = $1 - \text{Probability of no one solving it}$.

$$\therefore P(A_1 \cup A_2 \cup A_3 \cup A_4) = 1 - P(\bar{A}_1 \cap \bar{A}_2 \cap \bar{A}_3 \cap \bar{A}_4)$$

$$= 1 - P(\bar{A}_1)P(\bar{A}_2)P(\bar{A}_3)P(\bar{A}_4)$$

$$= 1 - [1 - P(A_1)][1 - P(A_2)][1 - P(A_3)][1 - P(A_4)]$$

$$= 1 - (1 - \frac{1}{2})(1 - \frac{1}{3})(1 - \frac{1}{4})(1 - \frac{1}{4}) = \frac{13}{16}$$

Problem 14. Mr.X and Mr.Y throw alternatively with a pair of ordinary dice. Mr.X wins if he throws 6 before Mr.Y throws 7 and Mr.Y wins if he throws 7 before Mr.X throws 6. If Mr.X begins the game show that the chance of his winning is $\frac{30}{61}$.

Solution. Let A denote the event of Mr.X throwing 6.

Let B denote the event of Mr.Y throwing 7.

Since 6 can occur in one of the given ways (1, 5), (5, 1), (2, 4), (4, 2) and (3, 3) we have $P(A) = \frac{5}{36}$.

Similarly $P(B) = \frac{6}{36} = \frac{1}{6}$. (verify)

$\therefore P(\bar{A}) = \frac{31}{36}$ and $P(\bar{B}) = \frac{5}{6}$.

Mr.X starts the game. Hence he has the following mutually exclusive ways of winning the game.

A happens (or) $\bar{A} \cap \bar{B} \cap A$ happens (or) $\bar{A} \cap \bar{B} \cap \bar{A} \cap \bar{B} \cap A$ happens etc.

$\therefore$ Probability of Mr.X winning the game

$$\begin{aligned} &= P(A) + P(\bar{A} \cap \bar{B} \cap A) + P(\bar{A} \cap \bar{B} \cap \bar{A} \cap \bar{B} \cap A) + \dots \\ &= P(A) + P(\bar{A})P(\bar{B})P(A) + P(\bar{A})P(\bar{B})P(\bar{A})P(\bar{B})P(A) + \dots \\ &= \frac{5}{36} + \left(\frac{31}{36} \times \frac{5}{6} \times \frac{5}{36}\right) + \left(\frac{31}{36} \times \frac{5}{6} \times \frac{31}{36} \times \frac{5}{6} \times \frac{5}{36}\right) + \dots \\ &= \frac{5}{36} + \left(\frac{5}{36}\right) \cdot \left(\frac{5}{6} \times \frac{31}{36}\right) + \left(\frac{5}{36}\right) \left(\frac{5}{6} \times \frac{31}{36}\right)^2 + \dots \\ &= \frac{\frac{5}{36}}{1 - \left(\frac{5 \times 31}{6 \times 36}\right)} = \frac{30}{61} \end{aligned}$$

Problem 15. X, Y, Z in order toss a fair coin. The first one to throw a head wins. What are their respective chances of winning. Assume that the game is continued indefinitely.

Solution. Let A be an event of X throwing head:

Let B be an event of Y throwing head.

Let C be an event of Z throwing head.

Then $P(A) = P(B) = P(C) = \frac{1}{2}$.

$\therefore P(\bar{A}) = P(\bar{B}) = P(\bar{C}) = \frac{1}{2}$.

If X starts the game then he has the following mutually exclusive ways of winning the game.

A happens or $\bar{A} \cap \bar{B} \cap \bar{C} \cap A$ happens or $\bar{A} \cap \bar{B} \cap \bar{C} \cap A \cap \bar{B} \cap \bar{C} \cap A$ happens etc.

$\therefore$ Probability of X winning the game

$$\begin{aligned} &= P(A) + P(\bar{A} \cap \bar{B} \cap \bar{C} \cap A) + P(\bar{A} \cap \bar{B} \cap \bar{C} \cap A \cap \bar{B} \cap \bar{C} \cap A) + \dots \\ &= P(A) + P(\bar{A})P(\bar{B})P(\bar{C})P(A) + P(\bar{A})P(\bar{B})P(\bar{C})P(A)P(\bar{B})P(\bar{C})P(A) + \dots \\ &= \frac{1}{2} + \frac{1}{2^4} + \frac{1}{2^7} + \dots = \frac{\frac{1}{2}}{1 - (\frac{1}{2})^3} = \frac{\frac{1}{2}}{\frac{7}{8}} = \frac{4}{7} \end{aligned}$$

Similarly probability of Y winning the game

$$\begin{aligned} &= P(\bar{A} \cap B) + P(\bar{A} \cap \bar{B} \cap \bar{C} \cap A \cap B) + \dots \\ &= \frac{1}{2^2} + \frac{1}{2^5} + \frac{1}{2^8} + \dots = \frac{(\frac{1}{2})^2}{1 - (\frac{1}{2})^3} = \frac{\frac{1}{4}}{\frac{7}{8}} = \frac{2}{7} \end{aligned}$$

Probability of Z winning the game

$$\begin{aligned} &= 1 - [\text{Probability of } X \text{ winning} + \text{Probability of } Y \text{ winning}] \\ &= 1 - \left(\frac{4}{7} + \frac{2}{7}\right) = \frac{1}{7} \end{aligned}$$

Problem 16. The probability that India wins a cricket test match against West Indies is known to be $\frac{2}{5}$. If India and West Indies play 3 test matches what is the probability that (i) India will loose all the three test matches. (ii) India will win atleast one test match (iii) India will win all the test matches (iv) India will win atmost one match.

Solution. Let A, B, C denote the events that India wins the first, second, third test match against West Indies respectively.

$$\text{Then } P(A) = P(B) = P(C) = \frac{2}{5}.$$

$$\text{Hence } P(\bar{A}) = P(\bar{B}) = P(\bar{C}) = \frac{3}{5}.$$

We note that the three events are independent events.

(i) Probability of India loosing all the three matches.

$$= P(\bar{A} \cap \bar{B} \cap \bar{C}) = P(\bar{A}) P(\bar{B}) P(\bar{C}) = \frac{3}{5} \times \frac{3}{5} \times \frac{3}{5} = \frac{27}{125}$$

(ii) Probability of India winning atleast one match

$$= 1 - (\text{Probability of loosing all the 3 matches})$$

$$= 1 - P(\bar{A} \cap \bar{B} \cap \bar{C}) = 1 - \frac{27}{125} = \frac{98}{125}.$$

(iii) Probability of India winning all the 3 matches.

$$= P(A \cap B \cap C)$$

$$= P(A) P(B) P(C) = \frac{2}{5} \times \frac{2}{5} \times \frac{2}{5} = \frac{8}{125}$$

(iv) Probability of India winning atmost one test match

$$= \text{Probability of loosing all matches} + \text{Probability of winning one match.}$$

$$= P(\bar{A} \cap \bar{B} \cap \bar{C}) + P(A \cap \bar{B} \cap \bar{C}) + P(\bar{A} \cap B \cap \bar{C}) + P(\bar{A} \cap \bar{B} \cap C)$$

$$= \frac{27}{125} + P(A) P(\bar{B}) P(\bar{C}) + P(\bar{A}) P(B) P(\bar{C}) + P(\bar{A}) P(\bar{B}) P(C)$$

$$= \frac{27}{125} + \left(\frac{2}{5} \times \frac{3}{5} \times \frac{3}{5}\right) + \left(\frac{3}{5} \times \frac{2}{5} \times \frac{3}{5}\right) + \left(\frac{3}{5} \times \frac{3}{5} \times \frac{2}{5}\right) = \frac{81}{125} \cdot (\text{verify})$$

Problem 17. A bag contains 5 red and 3 black balls and a second bag 4 red and 5 black balls. One of the bag is selected at random and a draw of 2 balls is made from it. What is the probability that one of them is red and the other is black.

Solution. Let A_1 and A_2 denote the event of selecting the first bag and second bag respectively.

$\therefore P(A_1) = 1/2 = P(A_2)$ and A_1 and A_2 are mutually exclusive events so that $A_1 \cup A_2 = S$ (sample space)

Let B denote the event of selecting one red and one black ball.

$$\text{Then } P(B|A_1) = \frac{5 \times 3}{8C_2} = \frac{15}{28} \text{ and } P(B|A_2) = \frac{4 \times 5}{9C_2} = \frac{5}{9}$$

$$\begin{aligned} \text{The required probability} &= P(A_1)P(B|A_1) + P(A_2)P(B|A_2) \\ &= \frac{1}{2} \cdot \frac{15}{28} + \frac{1}{2} \cdot \frac{5}{9} = \frac{15}{56} + \frac{5}{18} = \frac{275}{504} = 0.55 \end{aligned}$$

Problem 18. In a state election in 1995 there were 3 major political parties X, Y, Z fighting for the claim of chiefministership. The chance of winning the election of the 3 parties are in the ratio $1 : 2 : 3$ respectively. The probability that the party X if selected, will introduce total prohibition in that state is $1/2$. The probability that the party Y if selected, will introduce total prohibition in the state is $1/4$ and the probability that the party Z if selected, will introduce total prohibition in the state is $3/4$. What is the probability that there will be a total prohibition in the state after the election in 1995.

Solution. Let A_1, A_2 and A_3 be the events that parties X, Y and Z elected to claim chiefministership respectively.

$$\text{Given } P(A_1) = 1/6; P(A_2) = 2/6 = 1/3; P(A_3) = 3/6 = 1/2.$$

Let B be the event that total prohibition will be introduced.

$$\text{Given } P(B|A_1) = 1/2; P(B|A_2) = 1/4; P(B|A_3) = 3/4.$$

The required probability

$$\begin{aligned} &= P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3) \\ &= (1/6 \times 1/2) + (1/3 \times 1/4) + (1/2 \times 3/4) = 1/12 + 1/12 + 3/8 = 13/24. \end{aligned}$$

Problem 19. One factory F_1 produces 1000 articles 20 of them being defective, second factory F_2 produces 4000 articles, 40 of them being defective and third factory F_3 produces 5000 articles, 50 of them being defective. All these articles are put in one stock pile. One of them is chosen and is found to be defective. What is the probability that it is from the factory F_1 .

Solution. Let A_1, A_2 and A_3 denote the event that the chosen article is produced by the factory F_1, F_2 and F_3 respectively.

$$\therefore P(A_1) = \frac{1000}{10000} = \frac{1}{10}; P(A_2) = \frac{4000}{10000} = \frac{2}{5}; P(A_3) = \frac{5000}{10000} = \frac{1}{2}$$

Let B denote the event that the article chosen is defective. Then

$$P(B|A_1) = \frac{20}{1000} = \frac{1}{50}; P(B|A_2) = \frac{40}{4000} = \frac{1}{100}; P(B|A_3) = \frac{50}{5000} = \frac{1}{100}$$

We have to find $P(A_1|B)$

$$\begin{aligned} \text{By Baye's theorem } P(A_1|B) &= \frac{P(A_1) P(B|A_1)}{\sum_{i=1}^3 P(A_i) P(B|A_i)} \\ &= \frac{\frac{1}{10} \times \frac{1}{50}}{(\frac{1}{10} \times \frac{1}{50}) + (\frac{2}{5} \times \frac{1}{100}) + (\frac{1}{2} \times \frac{1}{100})} \\ &= \frac{\frac{1}{500}}{\frac{1}{500} + \frac{1}{250} + \frac{1}{200}} = \frac{1}{500} \times \frac{1000}{11} = \frac{2}{11} \end{aligned}$$

Problem 20. The contents of 3 urns are

Urn I : 1 white 3 red 2 black balls

Urn II : 3 white 1 red 1 black balls

Urn III : 3 white 3 red 3 black balls

Two balls are chosen from a randomly selected urn. If the balls are 1 white and 1 red ball what is the probability that they come from Urn II.

Solution. Let A_1 denote the event that Urn I is chosen.

Let A_2 denote the event that Urn II is chosen.

Let A_3 denote the event that Urn III is chosen.

$$\therefore P(A_1) = P(A_2) = P(A_3) = \frac{1}{3}.$$

Let B denote the event that the balls drawn are 1 white and 1 red ball.

We need to find $P(A_2 | B)$.

$$\text{By Baye's theorem } P(A_2 | B) = \frac{P(A_2) P(B | A_2)}{\sum_{i=1}^3 P(A_i) P(B | A_i)}$$

We now find $P(B | A_i)$ for $i = 1, 2, 3$.

$P(B | A_1)$ = Probability of getting 1 W and 1 R balls in Urn I.

$$= \frac{1 \times 3}{6C_2} = \frac{3}{15} = \frac{1}{5}.$$

$$\text{Similarly } P(B | A_2) = \frac{3 \times 1}{5C_2} = \frac{3}{10}$$

$$P(B | A_3) = \frac{3 \times 3}{9C_2} = \frac{9}{36} = \frac{1}{4}.$$

$$\begin{aligned} \therefore P(A_2 | B) &= \frac{\frac{1}{3} \times \frac{3}{10}}{\frac{1}{3} \left(\frac{1}{5} + \frac{3}{10} + \frac{1}{4} \right)} = \frac{\frac{1}{10}}{\frac{1}{3} \left(\frac{15}{20} \right)} = \frac{\frac{1}{10}}{\frac{1}{4}} \\ &= \frac{2}{5}. \end{aligned}$$

Exercises

- When 2 coins are tossed write a sample space and determine the following events. (i) getting exactly 2 heads, (ii) getting atleast one head, (iii) getting atmost one head, (iv) getting atleast one tail, (v) getting atmost one tail, (vi) getting exactly 2 tails.

2. If two dice are rolled write the event A of getting the sum 7 and the event B getting the same number on the two dice. Also assigning uniform probability $1/36$ to each sample point find (i) $P(A)$ (ii) $P(B)$ (iii) $P(A \cap B)$ (iv) $P(A \cup B)$.

3. When 3 coins are tossed write the sample space and determine the following events. (i) getting exactly 2 heads, (ii) getting atleast 2 heads, (iii) getting one tail, (iv) getting atleast 2 tails, (v) getting head and tail alternatively.

4. 3 dice are rolled. What is the probability that there is atleast one 6.

5. Considering a pack of cards write (i) the event of drawing a diamond card (ii) the event of drawing an ace (iii) find the probability of the above two events (iv) find the probability of getting a diamond and an ace (v) find the probability of getting a spade or ace.

6. If three cards are drawn from a pack of 52 cards (i) what is the probability that all three are queens (ii) what is the probability that they are a king, a queen and a knave.

7. Six cards are drawn at random from a pack of 52 cards. Find the probability that 3 will be red and 3 will be jack.

8. Find the probability of drawing an ace, a king, a queen and a knave in order from a pack of 52 cards in 4 consecutive draws the cards drawn being not replaced.

9. What is the chance that (i) a leap year selected at random will contain 53 sundays (ii) a non leap year should have 53 sundays.

10. Given $P(A) = a$, $P(B) = b$, $P(A \cap B) = c$. Find the following
(i) $P(A \cup B)$ (ii) $P(\overline{A \cap B})$ (iii) $P(\overline{A \cup B})$ (iv) $P(\overline{A \cup B})$ (v) $P(A | B)$
(vi) $P(B | A)$ (vii) $P(\overline{A} | \overline{B})$ (viii) $P(A \cup (\overline{A \cap B}))$

11. If $P(A) = 1/2$, $P(B) = 1/2$, $P(A \cup B) = 2/3$ find (i) $P(A \cap B)$
(ii) $P(\overline{A \cap B})$ (iii) $P(\overline{A \cup B})$

12. Let A and B be two events in a sample space such that $P(A) = 0.4$ and $P(B) = b$ and $P(A \cup B) = 0.7$. For what value of b
(i) A and B are mutually exclusive (ii) A and B are independent.

13. Let A and B be two events in a sample space. If $P(A \cup B) = 5/6$; $P(A \cap B) = 1/3$ and $P(\overline{B}) = 1/2$ find (i) $P(B)$ (ii) $P(A)$ (iii) Are $\overline{A}$ and B independent?

14. A, B, C are any three independent events such that $P(A) = 0.3$; $P(B) = 0.2$ and $P(C) = 0.1$. Find the probability of occurrence of atleast one of three events.

15. It is found that one type of defect occurs in a machine with a probability .05 and a second type of defect with a probability .02. What is the probability that an article produced by the machine (i) is defective (ii) does not have both types of defects.

16. If A and B are mutually exclusive events such that $P(B) = 2P(A)$ and $A \cup B = S$, show that $P(A) = \frac{1}{2}$.

17. If A, B and C are three events such that A and B are independent and $P(C) = 0$ prove that A, B, C are independent events.

18. If A and B are two exclusive disjoint events show that $P(A) \leq P(\bar{B})$.

19. If A and B are two events and $P(B) \neq 1$ prove that $P(A|B) = \frac{P(A) - P(A \cap B)}{1 - P(B)}$ and hence prove $P(A \cap B) \geq P(A) + P(B) - 1$.

20. Prove that if $P(A|B) < P(A)$ then $P(B|A) < P(B)$.

21. Let A and B be two events such that $P(A) = \frac{3}{4}$ and $P(B) = \frac{5}{8}$ show that (i) $P(A \cup B) \geq \frac{3}{4}$ (ii) $\frac{3}{8} \leq P(A \cap B) \leq \frac{5}{8}$.

22. A, B, C in order cut a pack of cards replacing them after each cut. Find their respective chances of cutting a heart.

23. A, B, C, D in order cut a pack of cards successively replacing them after each cut. Find their respective chances of cutting a spade.

24. The chance of A solving a problem is $\frac{3}{7}$ and B solving the same problem is $\frac{7}{12}$. Find the probability that the problem is solved if they both try independently.

25. A and B take turn in throwing two dice. The first to throw 9 is awarded the prize. Show that their respective chances are in the ratio 9 : 8.

26. A and B alternately cut a pack of cards and the pack is shuffled after each cut. If A starts and the game is continued until one cuts a diamond what are the respective chances of A and B first cutting a diamond.

27. A can hit a target four times in five shots; B three times in four shots and C twice in three shots. They each fire once at the same target. What is the probability that atleast two shots hit the target.

28. In a shooting test the probabilities of hitting the target are $\frac{1}{2}$ for A, $\frac{2}{3}$ for B and $\frac{3}{4}$ for C. If all of them fire at the same target find the probabilities that (i) only one of them hits the target (ii) atleast one of them hits the target.

29. Three Urns have the following composition of balls.

Urn I 1 white + 2 black

Urn II 2 white + 1 black

Urn III 2 white + 2 black

One of the urns is selected and a ball is drawn. It turns out to be white. Find the probability that it came from the Urn III.

30. In the coming year there will be three candidates Dr.Sing, Dr.Anand and Dr.Seelan for the position of principalship in a well established coeducation college whose chances of getting appointment are in the ratio 4 : 2 : 3 respectively. The probability that Dr.Sing if appointed will introduce sex education in the college is 0.3. The probability of Dr.Anand and Dr.Seelan doing the same are respectively 0.5 and 0.8. Find the probability that sex education is introduced in the college next year.

31. In a certain college the geographical distribution of lady students are as follows : 50% come from the East; 30% come from the Midwest; 20% come from the Fareast. The following percentage of lady students have the habit of smoking: 80% of the Eastern ladies; 60% of the Midwest ladies; 40% of the Fareast ladies. Find the probability that a lady student who has the habit of smoking from the East.

32. The contents of 3 urns are as follows

Urn I 1 White + 2 Black + 3 Red balls

Urn II 2 White + 1 Black + 1 Red balls

Urn III 4 White + 5 Black + 3 Red balls

One urn is selected at random and two balls are drawn. They happen to be white and red. Find the probability that they came from urn II or urn III.

